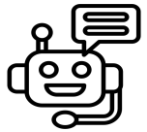
Workflow Development with n8n

Class seven

Types of AI-Powered Solutions

- Workflow 
  - Automated **process pipeline**
- Intelligent Chatbot 
  - The system could perform **iterative queries and analysis** based on the end user queries
- Agentic Assistant 
  - It will understand, plan and execute the tasks **autonomously**



PDF Summarization Engine

Resume Summarization engine

Types of nodes in n8n-1/2

Category	Purpose	Examples of Nodes
Trigger Nodes	Start a workflow when something happens	Webhook Trigger, Schedule Trigger (Cron), Gmail Trigger (new email), Telegram Trigger (new message)
App / Action Nodes	Perform operations with specific apps & services	Google Sheets (add row), Gmail (send email), Slack (post message), Notion (create page), Airtable (update record)
Data Transformation	Clean, reshape, or restructure data	Set, Move/Remove Fields, Rename Keys, Item Lists, Split In Batches
Logic / Control Flow	Control workflow execution path	IF, Switch, Merge, Wait, Error Trigger
Function / Code	Add custom programming logic	Function (JavaScript), Function Item (per item JS), Code (JavaScript / Python)
AI node	AI reasoning & tool calling	OpenAI, Azure OpenAI**, AI Agent ,**Think Node, Embeddings
Utility Nodes	General-purpose helpers	**HTTP Request, Webhook**, Date & Time, Crypto, Spreadsheet File, File Operations
Database Nodes	Read/write structured data	MySQL, PostgreSQL, MongoDB, SQLite, Notion DB, Airtable
Messaging & Notifications	Communicate with users	Email (IMAP/SMTP/Gmail/Outlook), Slack, Teams, Telegram, Discord, Twilio (SMS/WhatsApp)

Types of nodes in n8n-2/2

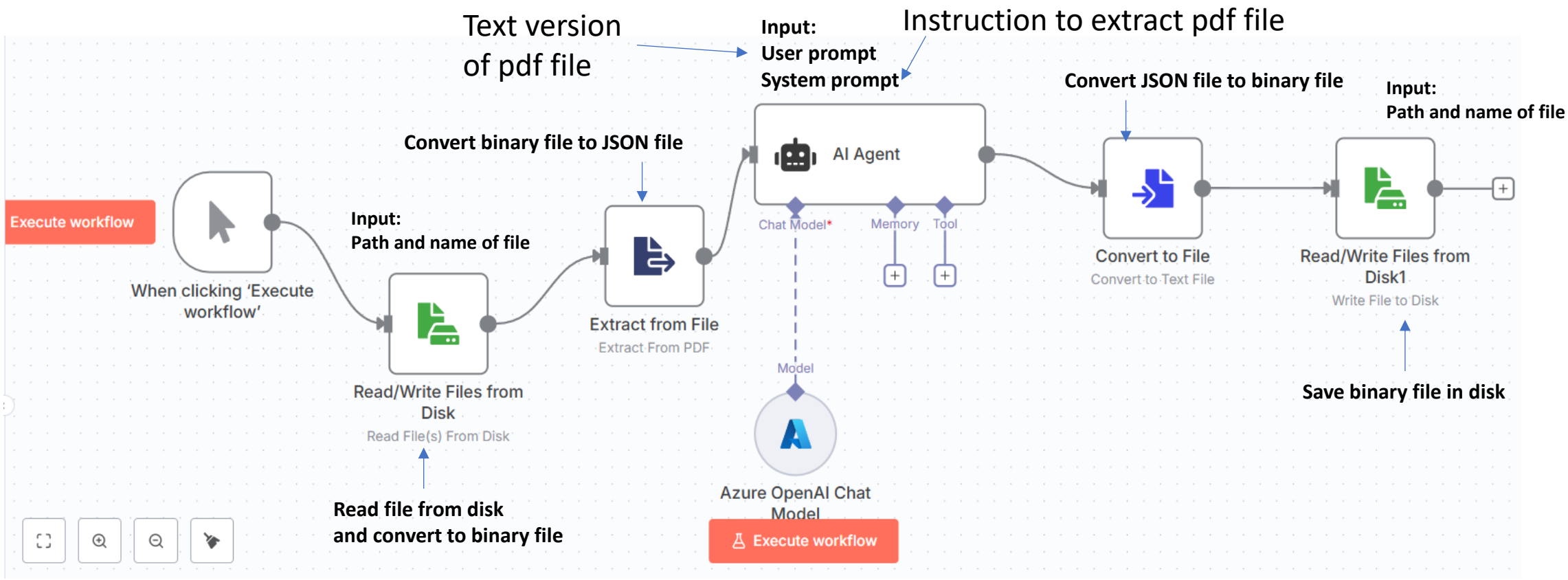
Node	
Read/Write files from Disk	Import files(e.g. pdf/csv) to/from binary
Extract from file	Convert binary file to JSON
Convert to file	Convert JSON data to binary file
Edit Fields	Create/edit content (JSON)
AI Agent	AI framework
Memory	Memory module used in AI framework
LLM	e.g. Azure and other LLM
Web Hook	To receive http requests from other server
Command Node	To execute command (e.g. execute bat file under Windows OS)
HTTP Request	To send HTTP request to other nodes/server/AI Agent
Wait	To pend and wait for further message (e.g. web hook) or condition (time expired)
IF/SWITCH	Condition branching
Application nodes	Google/Meta/Telegram/Whatsapp/etc



Case one

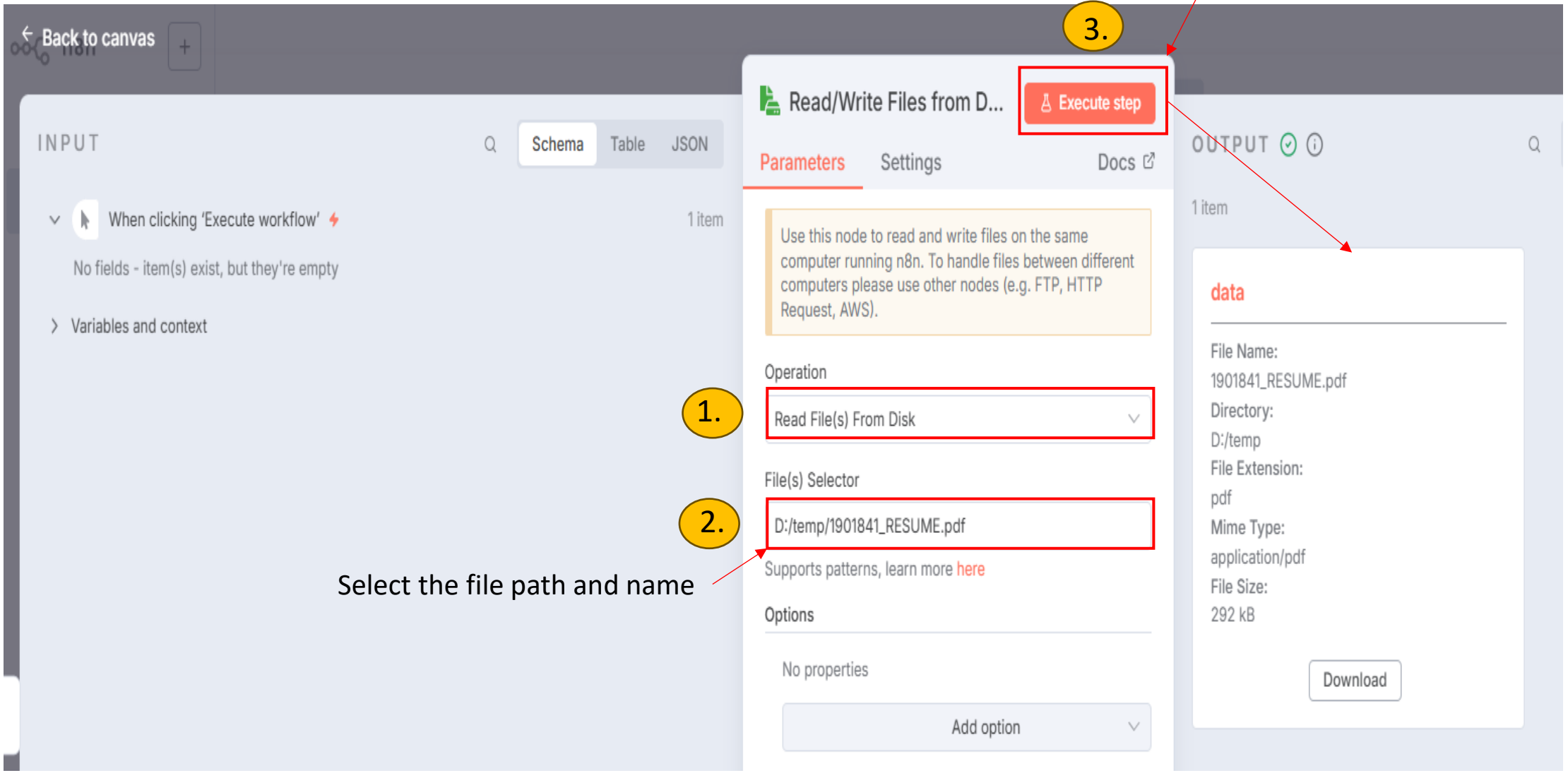
Single PDF file

Use Case
Resume summarization and analysis engine



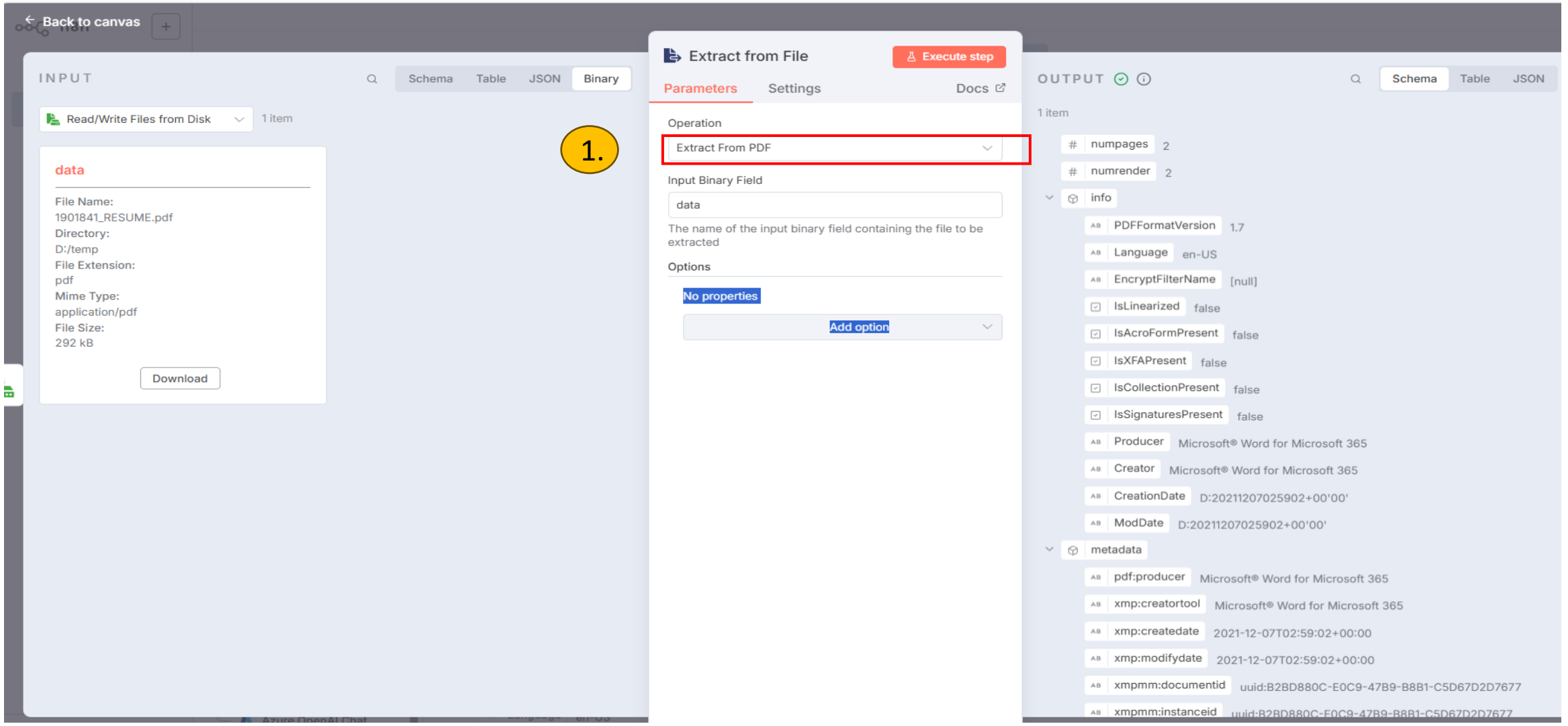
1. Read/Write Files from Disk

Click this one and see the execution result



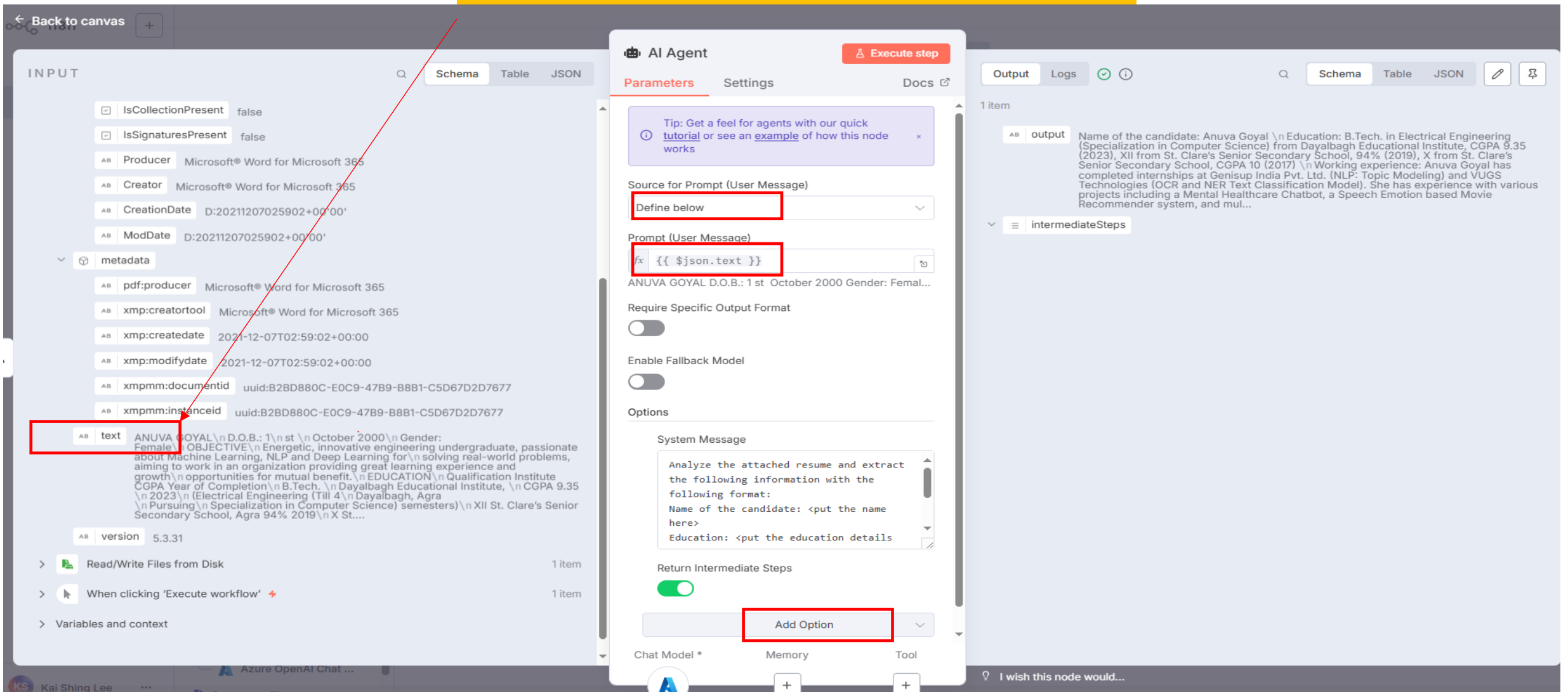
Select the file path and name

2. Extract from file



3. AI Agent

You could simply drag and drop from here to include the variable {{$json.text}}



3. AI Agent
System prompt

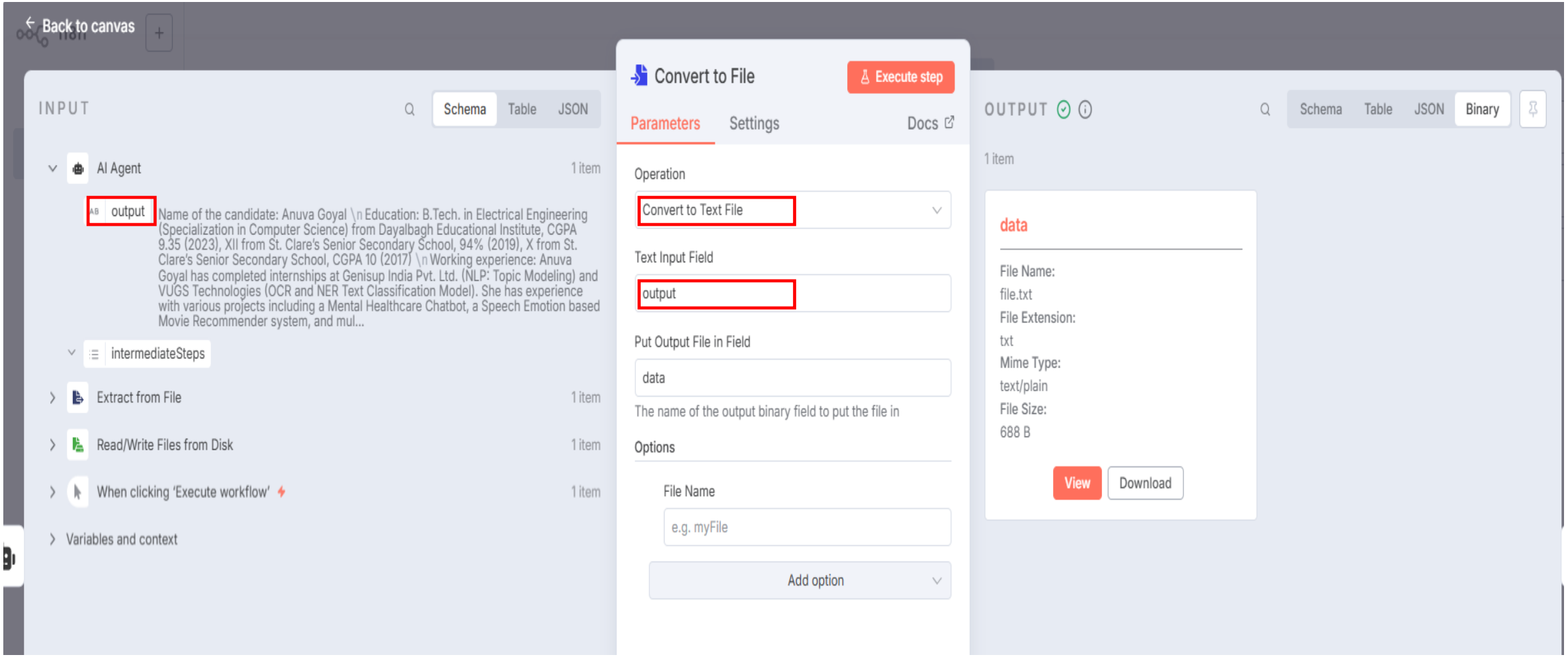
Analyze the attached resume and extract the following information with the following format:

Name of the candidate: <put the name here>

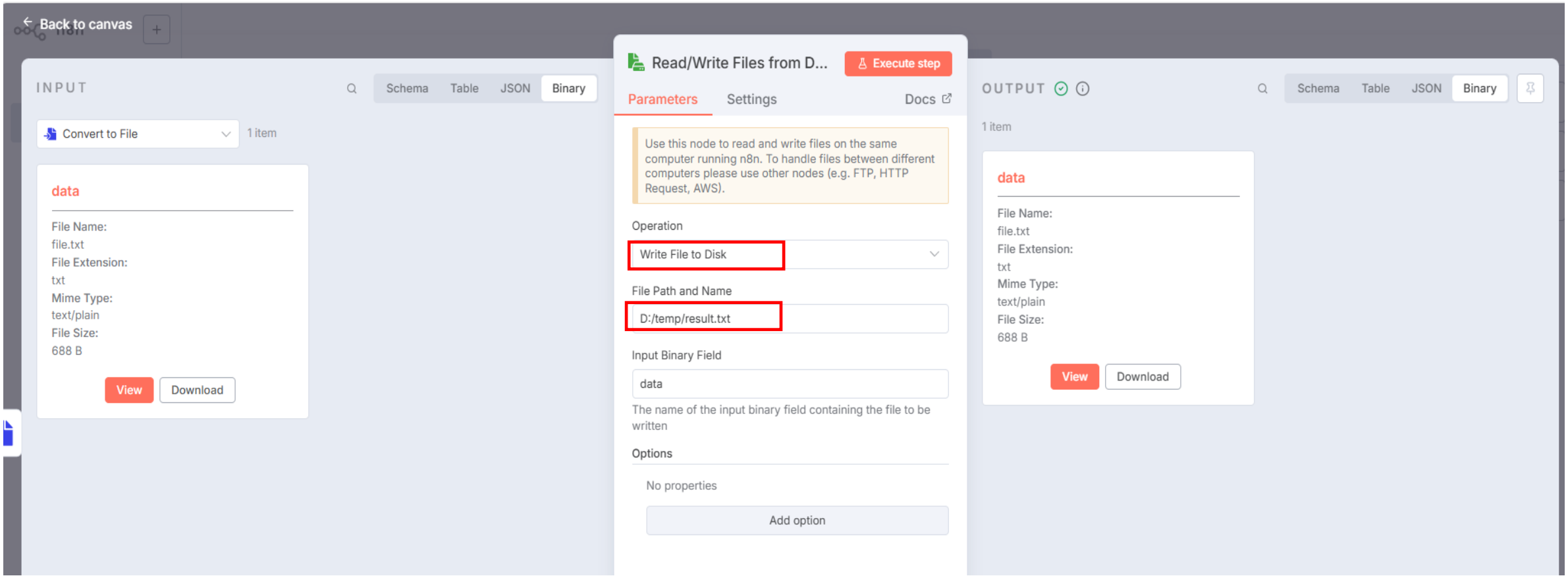
Education: <put the education details here>

Working experience: <put the summarize of the working experience>

4. Convert to file



5. Read/Write Files from Disk



Setup Google API

Types of Google API

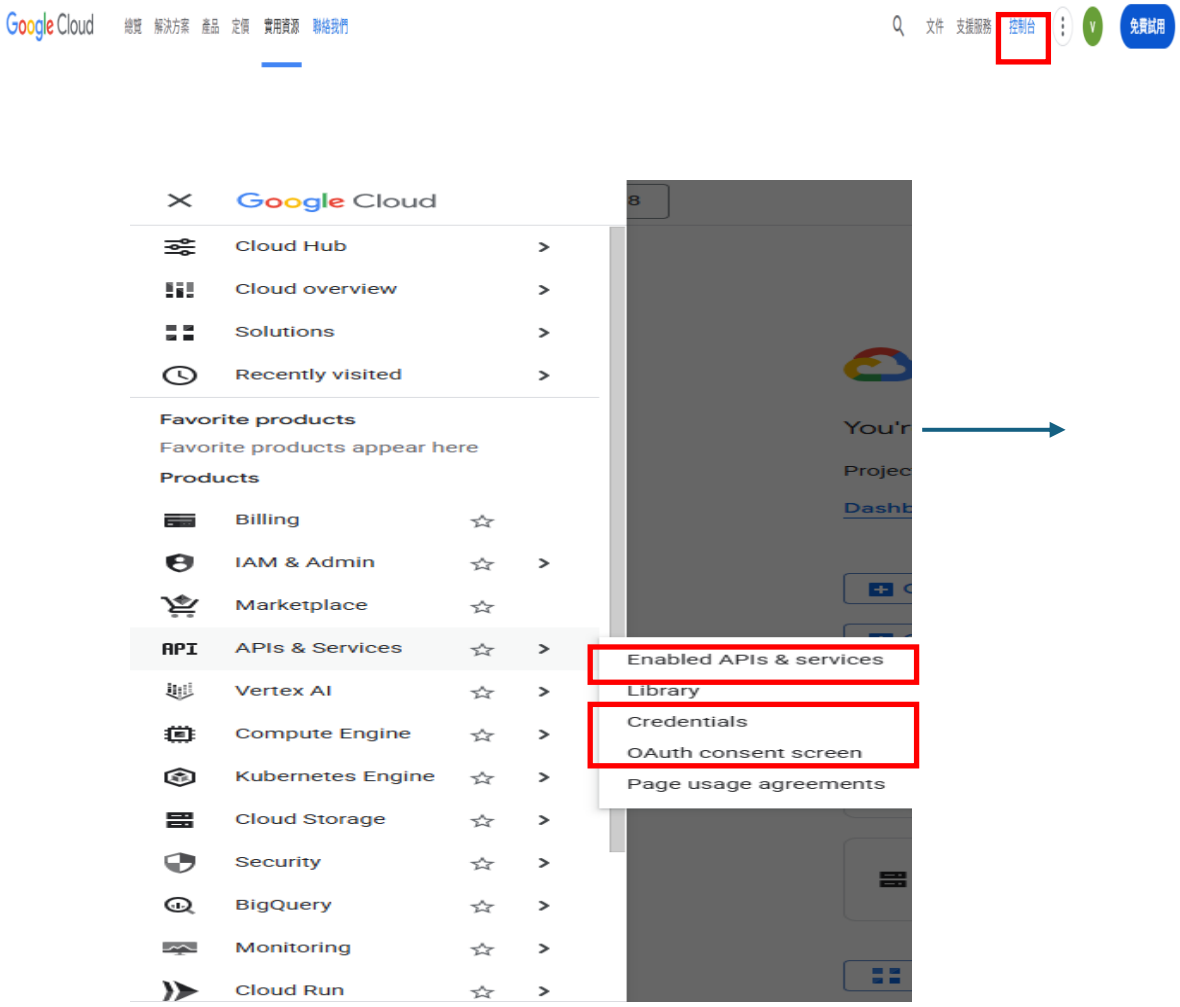
🚦 Quick comparison

Feature	OAuth (User-based)	Service Account (Robot-based)
Represents	A human user account	A non-human account (bot)
Consent needed	Yes (user clicks "Allow")	No (uses key)
Works with "My Drive"	✅ Yes	❌ No (unless shared)
Works with Gmail	✅ Yes	❌ No (unless domain delegation)
Token lifetime	Needs refresh, stable in Production mode	Stable, key-based
Best for	Personal assistants, Gmail, Calendar, Drive automation	Server-to-server, backend jobs, shared drive processing

Need to share the Resource (e.g. google sheet) with the bot

Setup of OAuth (User Based)

Overall Steps (OAuth)



Create new project

Specify the required services

Gmail/Google Drive etc

Configure the OAuth Consent Screen

URL callback from Dify/n8n

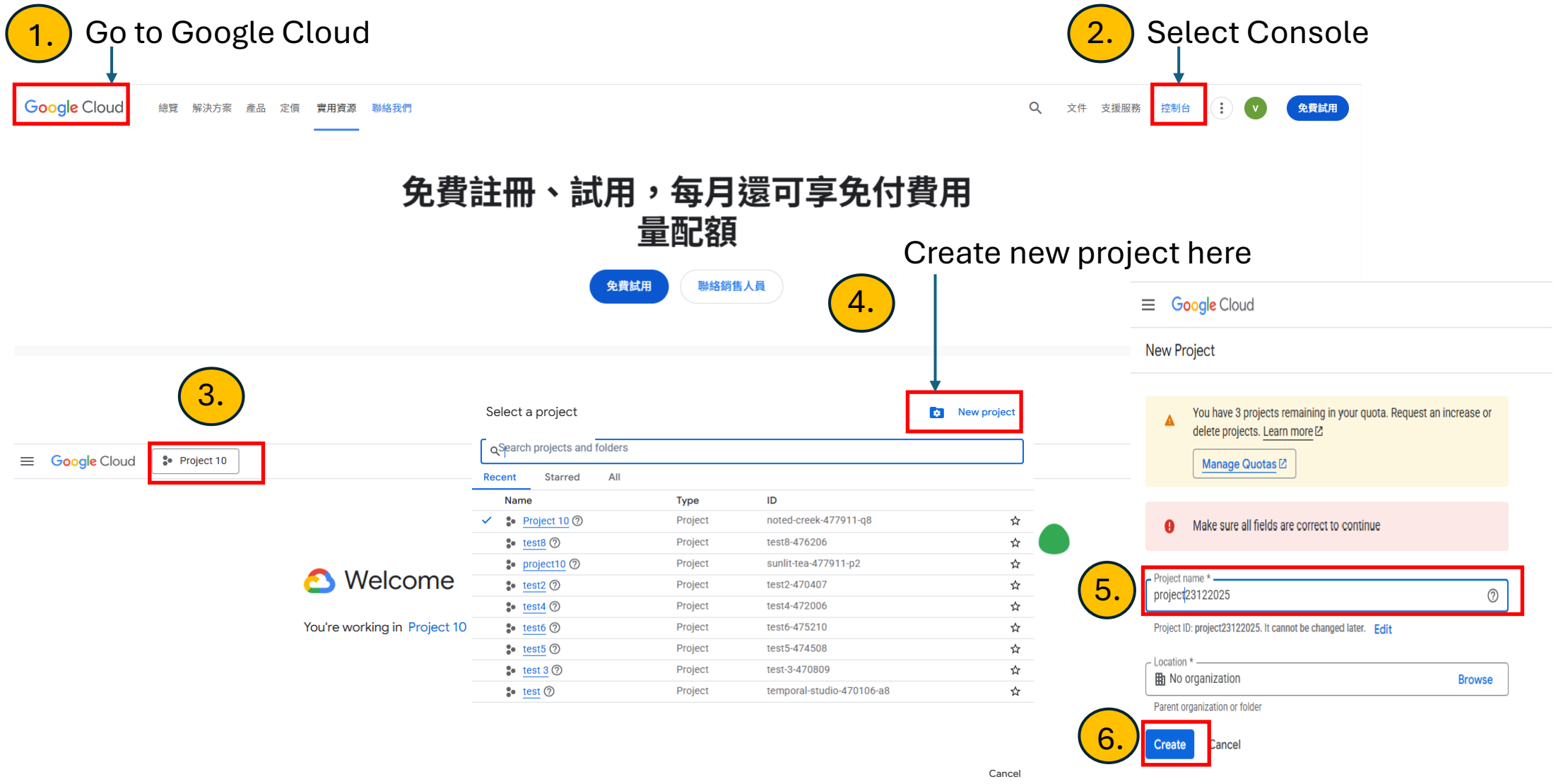
Configure the OAuth Client

Dify/n8n

Client ID
Secret

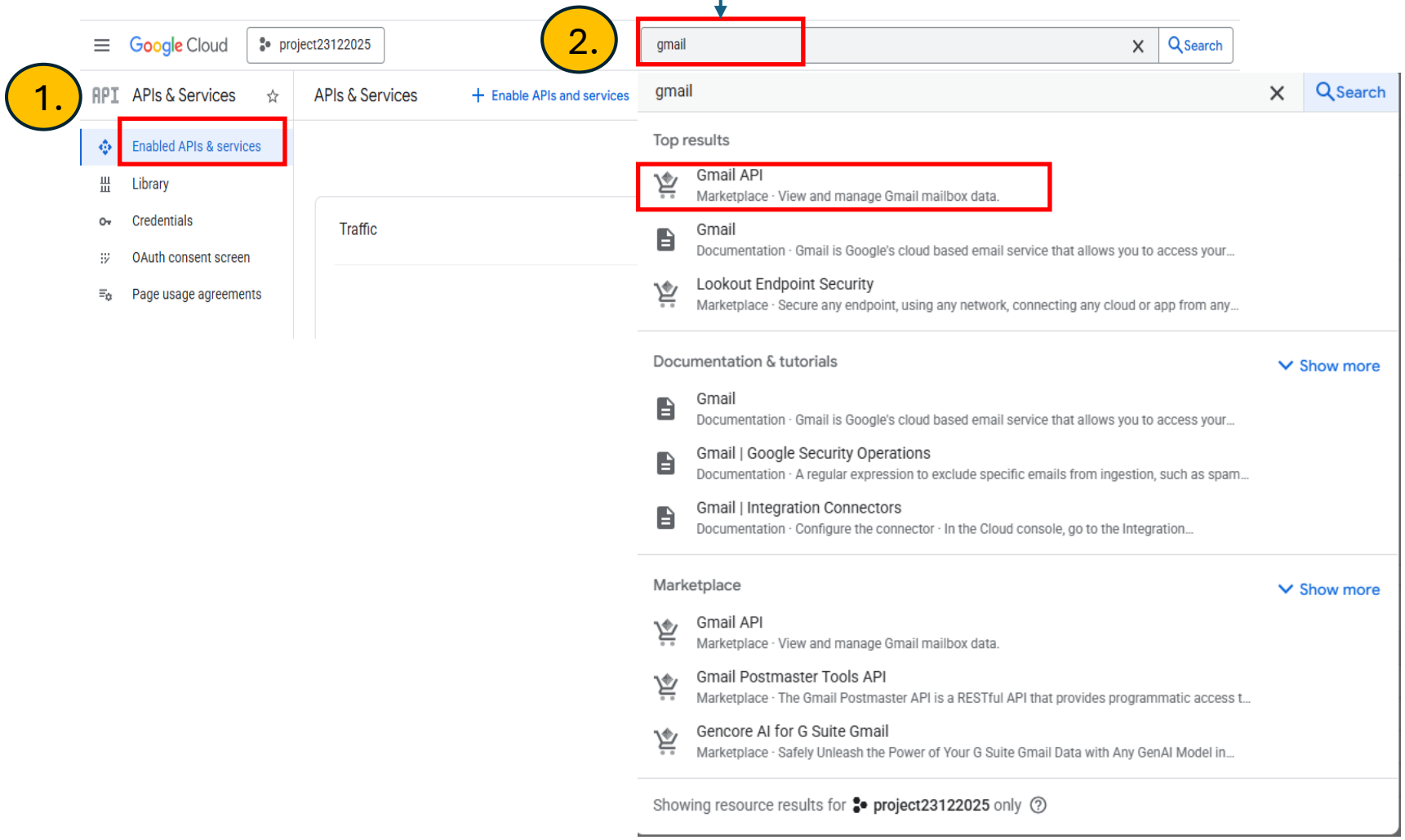
Fill in other information (test users)

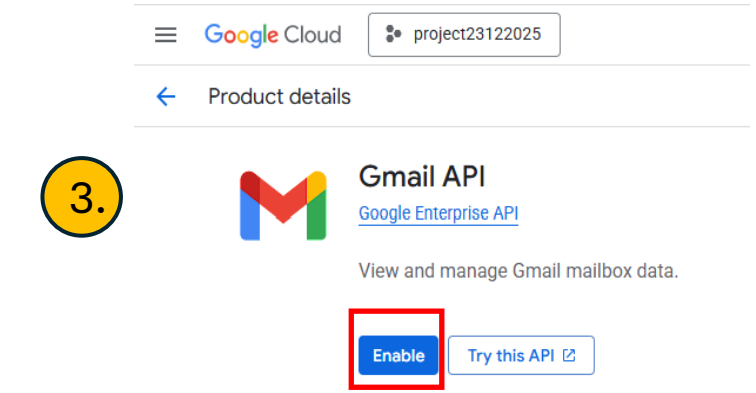
Create New Project



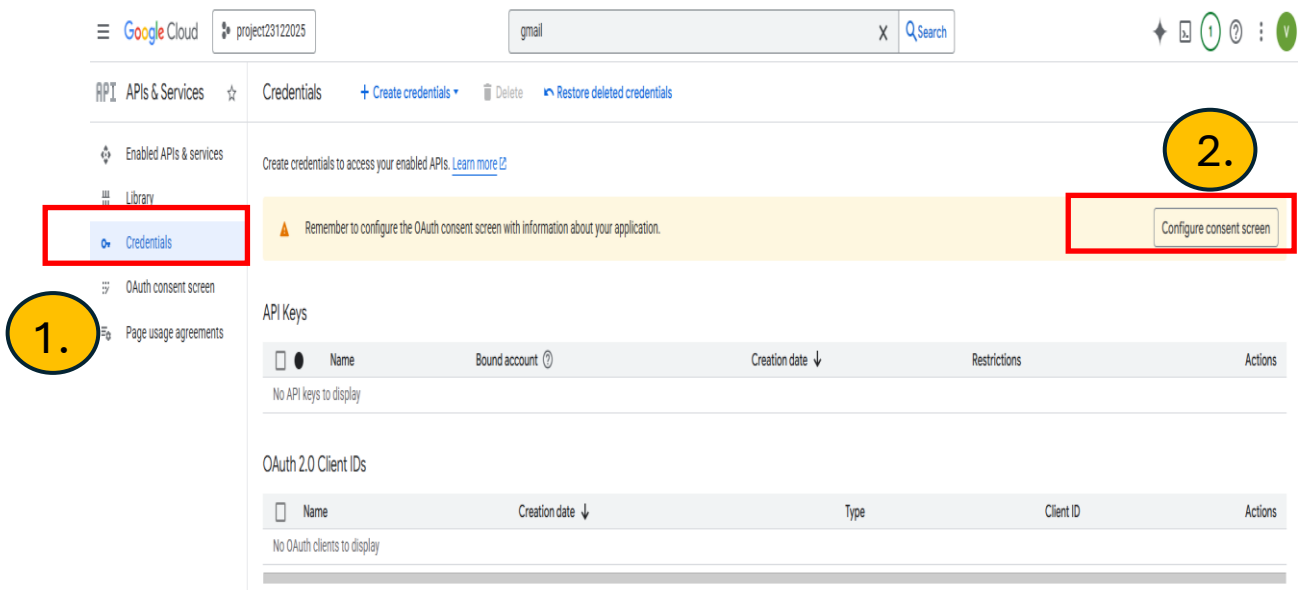
Enable API & Services

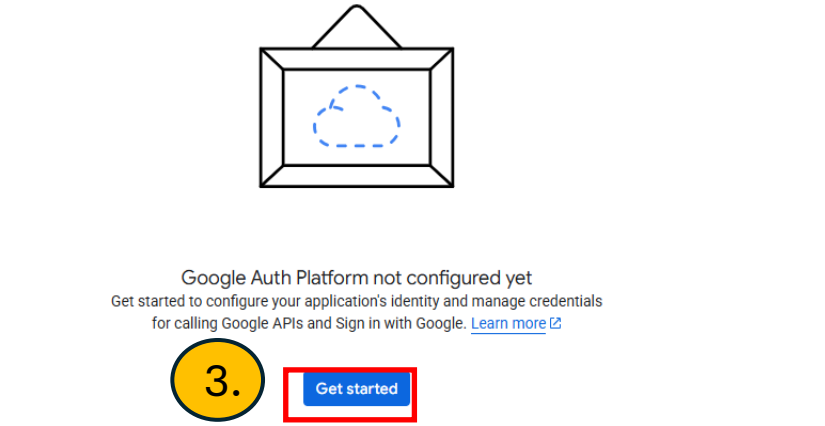
Example: Enable Gmail service

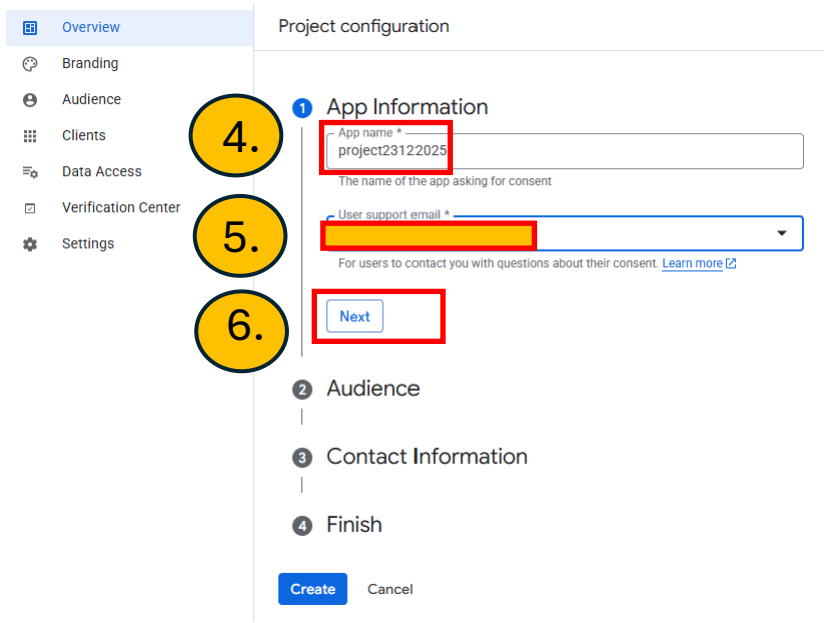




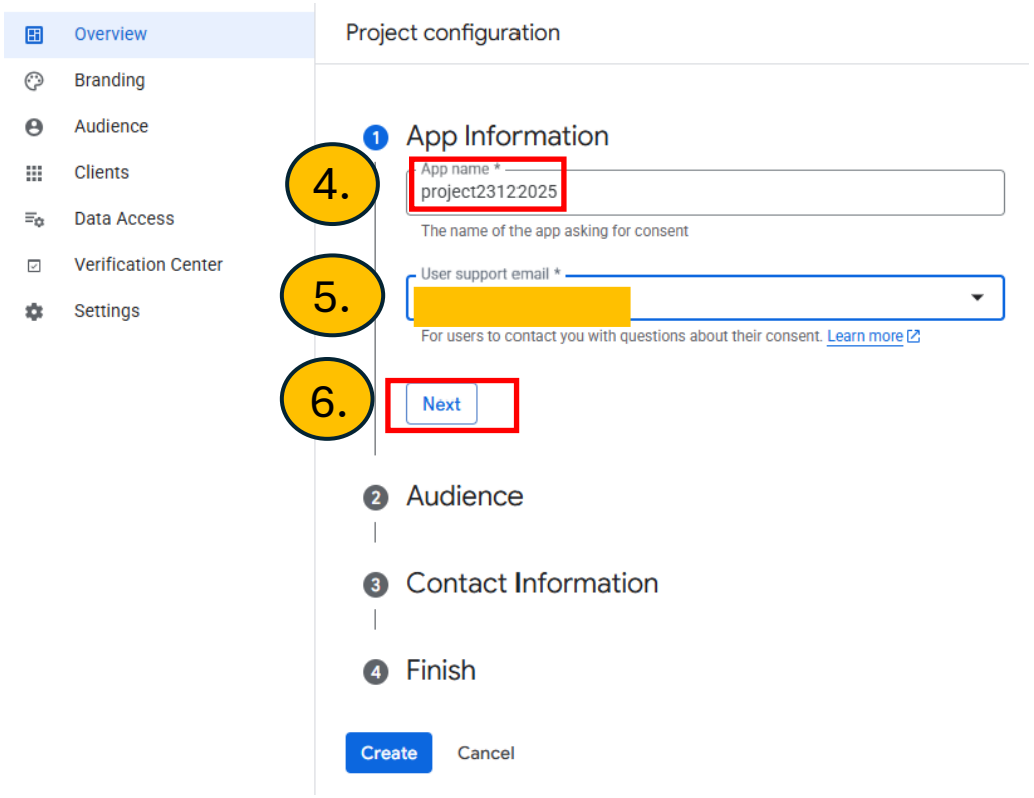
Setup Consent Screen-1/3

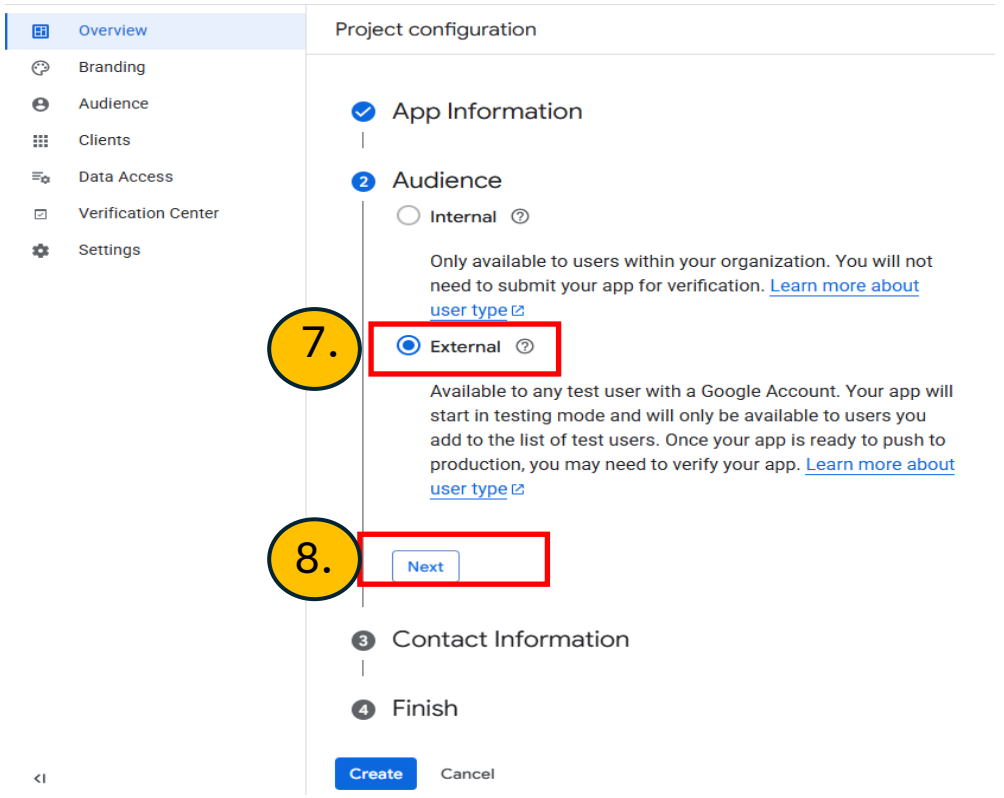




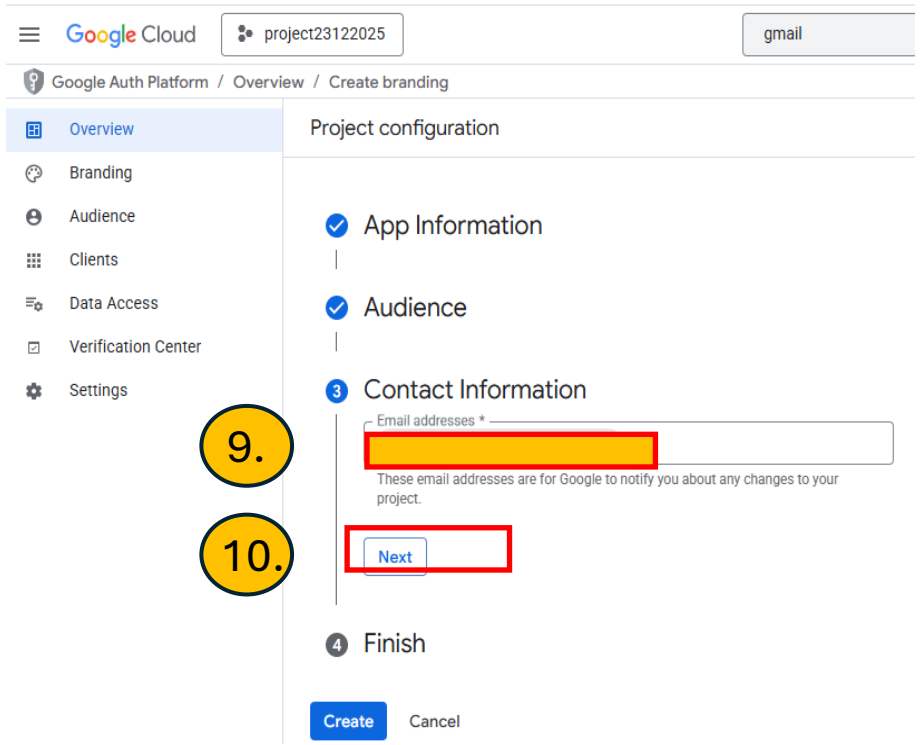


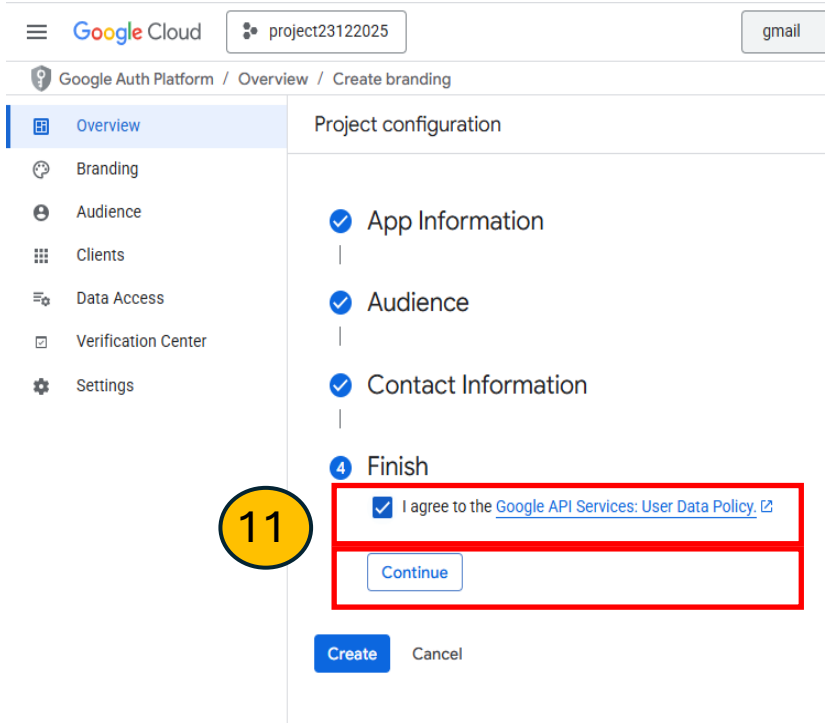
Setup Consent Screen-2/3



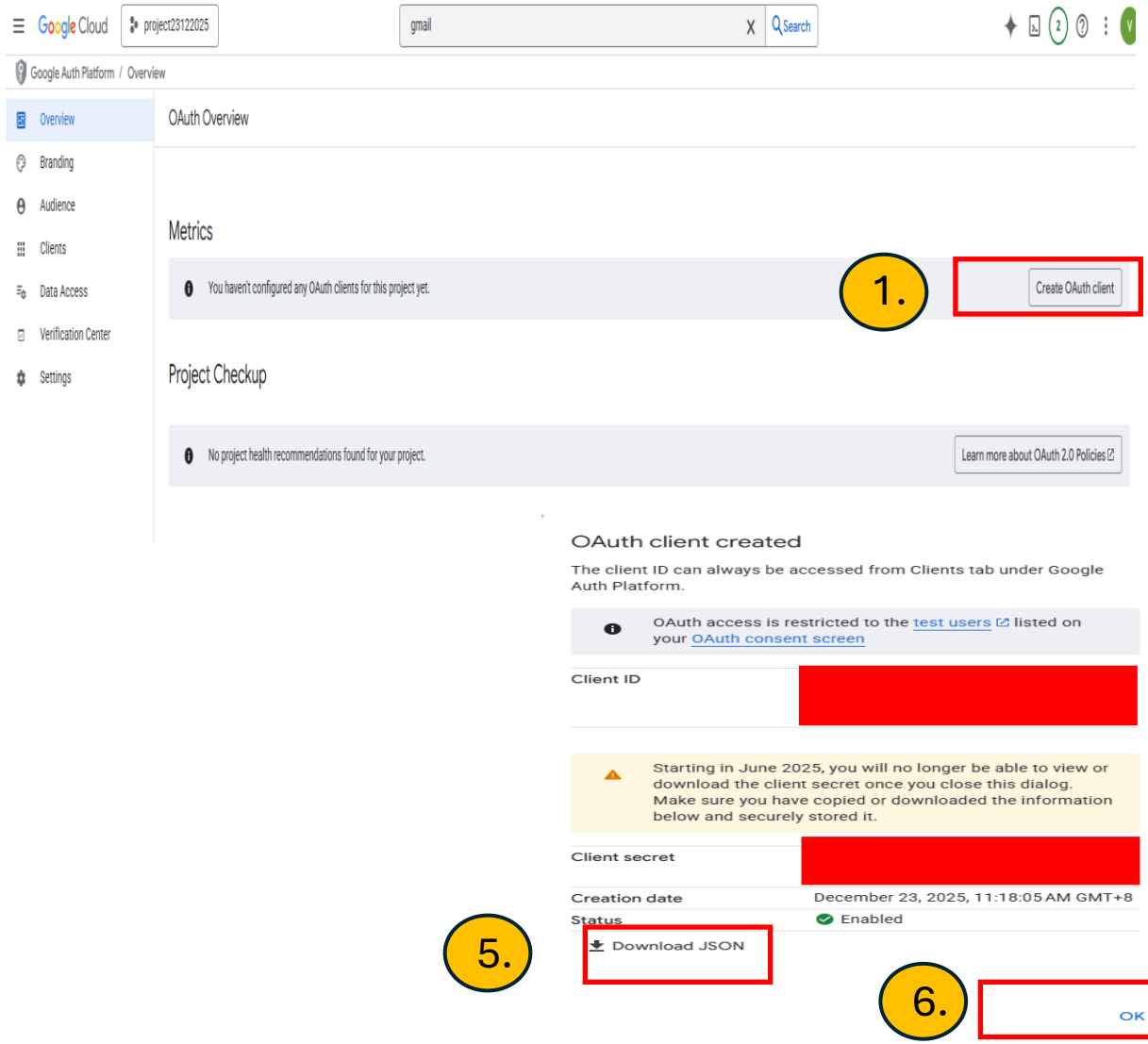


Setup Consent Screen-3/3

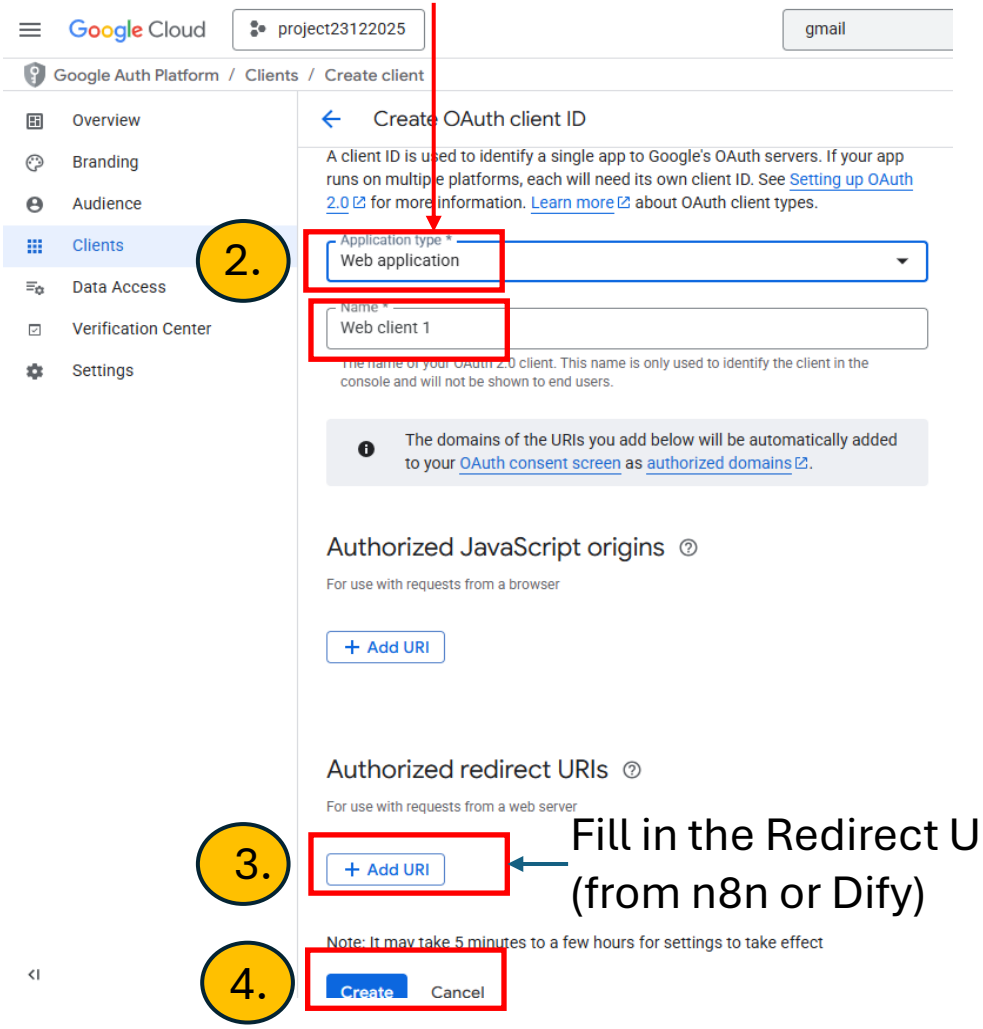




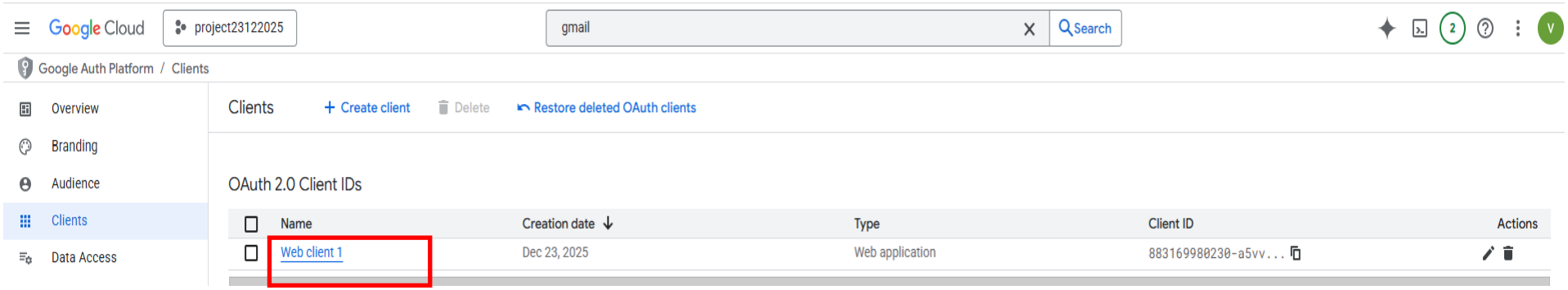
Configure OAuth Client-1/2



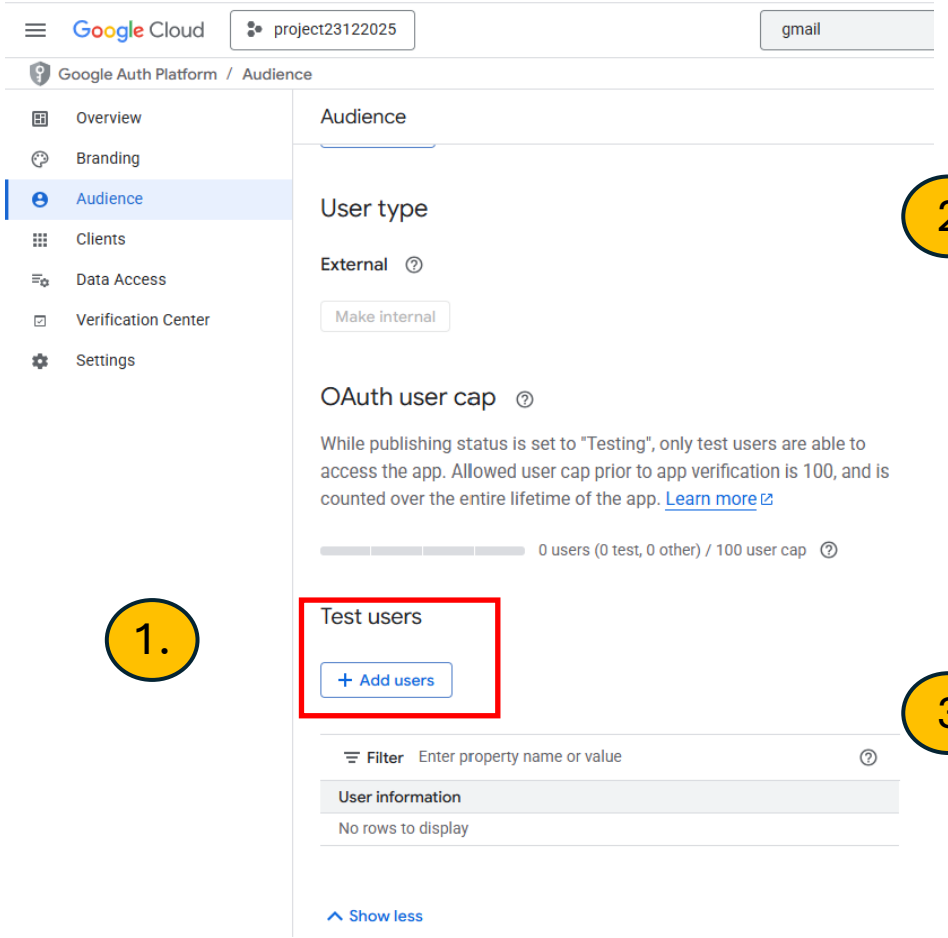
Select Web Application



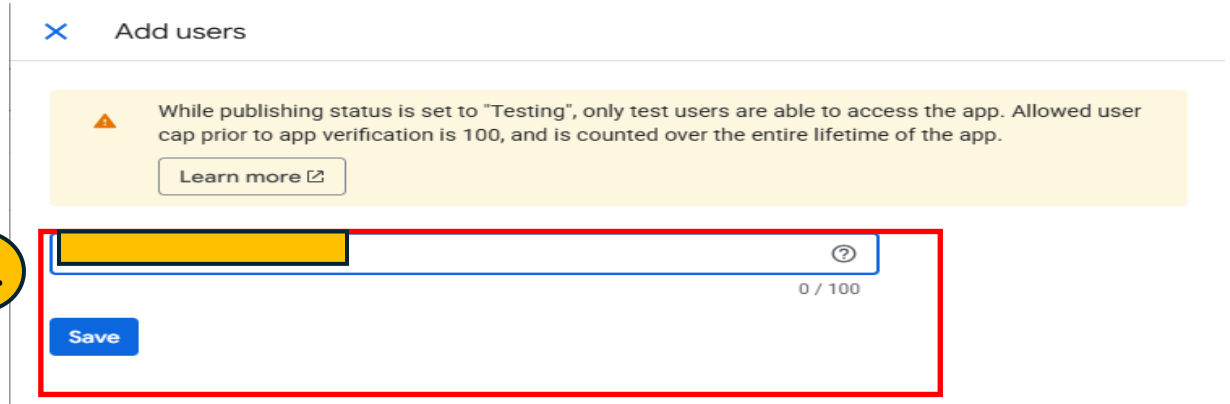
Configure OAuth Client-2/2



Create Test Users



1.



2.



3.